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Unifying Statistical Mechanics and Machine Learning: ELBO Topologies and Bifurcations in Markov Random Fields

FINAL WORK OF
METHODS AND MODELS FOR STATISTICAL MECHANICS

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Academic year:
2025-2026

Abstract: Variational Inference (VI) has emerged as a computationally efficient alternative to Markov Chain Monte Carlo (MCMC) methods for approximating intractable posterior distributions, primarily relying on the Coordinate Ascent Variational Inference (CAVI) algorithm. However, while the statistical properties of VI estimators are extensively documented, the algorithmic convergence guarantees of CAVI remain poorly understood, especially in non convex regimes. This report formalizes a connection between statistical mechanics and machine learning optimization by analyzing the CAVI algorithm as a discrete dynamical system. Using the two-node Ising model as a foundational Markov Random Field (MRF), we establish a mathematical correspondence between physical phase transitions and algorithmic bifurcations. We derive analytically that within the high temperature Dobrushin regime, the Evidence Lower Bound (ELBO) is globally strictly concave, guaranteeing global convergence to the unique variational optimum. Below the critical threshold, the ELBO landscape loses its global concavity and develops multiple stationary points. Through linear stability analysis, we prove that the symmetric stationary point transforms into an unstable hyperbolic saddle, trapping sequential CAVI in suboptimal local modes via a supercritical pitchfork bifurcation and driving parallel CAVI into non convergent oscillatory dynamics. We conclude by showing how the Edwards-Sokal coupling, which acts as a form of auxiliary latent variable augmentation, modifies the ELBO geometry to expand the convergence domain, providing critical insights into the optimization limits and structural requirements of modern Variational Inference.

Key-words: Variational Inference, Coordinate Ascent Variational Inference (CAVI), Evidence Lower Bound (ELBO), Dynamical Systems, Bifurcation Theory, Ising Model, Mean-Field Approximation

1. Introduction

Variational Inference (VI) represents a foundational framework for scalable Bayesian statistics and complex spatial modeling, reframing the computation of intractable posterior distributions as a deterministic optimization problem. The most widespread implementation of this framework is Mean-Field Variational Inference (MFVI), which restricts the variational family to the class of fully factorized product distributions. The optimization is typically executed by maximizing the Evidence Lower Bound (ELBO) via the Coordinate Ascent Variational Inference (CAVI) algorithm. Despite the empirical success of CAVI, its algorithmic stability in strong correlated, non convex regimes remains less understood. While classical convergence guarantees exist for log-concave distributions, characterizing how the geometry and dynamical stability of the optimization landscape depend on the interaction structure of the underlying probabilistic model remains an open theoretical question. Existing literature deeply investigates the statistical optimality of the resulting estimators, but frequently assumes that the optimization algorithm successfully navigates this complex topology to reach the global maximum of the ELBO. In highly parameterized, non conjugate spaces, the ELBO is non convex, raising critical questions regarding algorithmic stability and the reliability of the resulting variational estimates [3]. This report investigates the algorithmic boundaries of CAVI by formulating its update equations as an iterated, discrete time dynamical system. Using the two-node Ising model as a foundational Markov Random Field [5], we first derive the ELBO geometry and establish the formal monotonic convergence guarantees of the sequential updates. We then characterize the local stability of the ELBO stationary points through linear stability analysis. Specifically, we apply bifurcation theory to determine the critical temperature thresholds where the mean-field approximation loses global concavity. We demonstrate analytically that crossing this threshold causes sequential coordinate descent to converge to biased local modes, while parallel block updates are driven into non convergent oscillatory dynamics. Furthermore, we evaluate how symmetry-breaking external fields, acting as informative Bayesian priors, alter this non convex topology and suppress these dynamical traps. Finally, we show how introducing auxiliary latent variables via the Edwards-Sokal coupling [1] modifies the optimization landscape, thereby offering a structural mechanism to preserve algorithmic stability in strongly coupled networks.

2. Historical Overview and Context

The foundational objective of Bayesian statistics is the computation of the posterior distribution, a task that relies on evaluating the marginal likelihood. Historically, the primary computational bottleneck in evaluating this normalizing constant was circumvented through stochastic sampling techniques, most notably Markov Chain Monte Carlo (MCMC) methods such as the Metropolis-Hastings algorithm and Gibbs sampling [4]. While MCMC provides asymptotic guarantees of recovering the true target distribution, its computational cost and slow mixing times in high-dimensional spaces render it infeasible for many modern applications. This limitation led to the adoption of deterministic optimization via Variational Inference (VI), which sacrifices exactness for significant gains in computational speed by finding the best approximation to the posterior from a specified family of tractable distributions [6]. These variational approximations originated in the statistical physics of spin glasses and magnetism [5]. The Ising model, originally designed to explain ferromagnetic phase transitions, provides the mathematical foundation for this framework. Although rooted in physics, the Ising topology, representing local dependencies across a discrete lattice, has been adapted as a core structure in modern spatial statistics and Machine Learning, with its minimal two-node formulation serving as the foundational unit for convergence analysis. In geostatistical frameworks, such as Kriging, spatial dependencies are modeled via parameterized covariance functions; the Ising model's interaction matrices serve an identical conceptual purpose, dictating local spatial autocorrelation structures where neighboring nodes mutually influence each other [2].

The application of mean-field theory from statistical mechanics to predictive modeling initially carried the assumption that Coordinate Ascent Variational Inference (CAVI) algorithms would smoothly converge to statistically optimal estimates. However, recent work in machine learning has identified

structural instabilities in these methods: in strongly correlated, non conjugate spaces, the Evidence Lower Bound (ELBO) is non convex [3]. This non convexity acts as a major optimization bottleneck, causing trajectories to become trapped in suboptimal local modes or exhibit infinite oscillatory dynamics. Understanding the geometric boundaries where these topological traps occur is therefore crucial for deploying reliable variational estimators in complex spatial models.

3. Mathematical Notation and Fundamental Definitions

Before analyzing the coordinate ascent dynamics, we establish the mathematical definitions required to map the statistical mechanics of the Ising model to the probabilistic framework of machine learning.

Definition 1 (Configuration / State Space $\mathcal{X}$):

Let V be a finite set of n nodes (or vertices) representing interacting variables. The configuration space $\mathcal{X}$ is the discrete hypercube defining all possible global states of the system, such that $x \in \mathcal{X} = \{-1, 1\}^n$.

Physical Intuition: Each x_u represents the magnetic spin of a particle, which points either "up" (+1) or "down" (-1).

Machine Learning Intuition: The space $\mathcal{X}$ represents all possible joint assignments of a binary Markov Random Field (MRF), where each node is a discrete random variable in a spatial network.

Definition 2 (Hamiltonian / Energy Function $H(x)$):

The Hamiltonian is a scalar function $H : \mathcal{X} \rightarrow \mathbb{R}$ that maps a global configuration x to its total energy. For a pairwise interacting system, it is defined as:

$$H(x) = - \sum_{u \sim v} J_{uv} x_u x_v - \sum_{u \in V} h_u x_u \quad (1)$$

where J_{uv} is a symmetric interaction matrix with a zero diagonal ($J_{uu} = 0$), and h_u is the external field vector.

Physical Intuition: At zero temperature, the system stabilizes at its global minimum energy configuration; at finite temperatures, the probability of occupying a state depends on its energy, with lower energy configurations remaining thermodynamically more probable.

Machine Learning Intuition: The Hamiltonian corresponds to the negative unnormalized log-posterior distribution. The interaction term J_{uv} encodes the spatial autocorrelation (acting as a matrix proxy), while the external field h_u acts as a node-specific prior bias.

Definition 3 (Partition Function $\mathcal{Z}$):

The partition function is the fundamental normalizing constant of the system, computed by summing the Boltzmann factor over the entire configuration space $\mathcal{X}$:

$$\mathcal{Z} = \sum_{x \in \mathcal{X}} \exp \{-\beta H(x)\} \quad (2)$$

where $\beta \geq 0$ is the inverse temperature parameter.

Physical Intuition: $\mathcal{Z}$ encodes the macroscopic thermodynamic properties of a system in equilibrium, while β determines the magnitude of thermal fluctuations.

Machine Learning Intuition: This constant represents the marginal likelihood (or evidence) in Bayesian inference. Because $|\mathcal{X}| = 2^n$, direct evaluation of $\mathcal{Z}$ scales exponentially with the number of variables, creating the computational bottleneck that necessitates variational approximations.

Definition 4 (Gibbs / Boltzmann Measure):

The Gibbs measure is the unique probability distribution $p(x)$ over the configuration space $\mathcal{X}$ that maximizes the information entropy of the system for a given expected energy:

$$p(x) = \frac{1}{\mathcal{Z}} \exp \{-\beta H(x)\} \quad (3)$$

Physical Intuition: This physical law dictates the probability of finding a system in a specific configuration x at thermal equilibrium.

Machine Learning Intuition: This equation represents the true, strongly coupled joint posterior distribution of the spatial network. The goal of Mean-Field Variational Inference is to approximate this distribution using a factorized family of independent marginals.

Definition 5 (Mean-Field Variational Family $\mathcal{F}_{MF}$):

The mean-field variational family is defined as the set of all probability distributions $q(x)$ over $\mathcal{X}$ that factorize completely into the product of independent marginal distributions for each variable:

$$\mathcal{F}_{MF} := \left\{ q(x) = \prod_{u \in V} q_u(x_u) \right\}$$

where each $q_u(x_u)$ is a valid probability measure over the discrete state space $\{-1, 1\}$.

Definition 6 (Evidence Lower Bound - ELBO):

Given an unnormalized target distribution $\tilde{p}(x) = \exp\{-\beta H(x)\}$ and its normalized counterpart $p(x) = \tilde{p}(x)/\mathcal{Z}$, the Evidence Lower Bound (ELBO) for a variational distribution $q(x)$ is a scalar objective function defined as:

$$\text{ELBO}(q) = \mathbb{E}_q[\log \tilde{p}(x)] - \mathbb{E}_q[\log q(x)]$$

Maximizing the ELBO with respect to $q(x)$ is mathematically equivalent to minimizing the Kullback-Leibler (KL) divergence $D_{KL}(q||p)$.

4. Theoretical Framework and The Ising MRF

As established in Section 3, the Ising model can be structured as a prototypical undirected probabilistic graphical model, formally known as a pairwise Markov Random Field (MRF). For a system of n binary variables over the vertex set V , where global configurations satisfy $x \in \{-1, 1\}^n$, the joint probability distribution is governed by the Boltzmann measure defined in (3):

$$p(x) = \frac{1}{\mathcal{Z}} \exp \left\{ \beta \sum_{u \sim v} J_{uv} x_u x_v + \beta \sum_{u \in V} h_u x_u \right\}, \quad (4)$$

where $\beta \geq 0$ is the inverse temperature controlling the stochasticity of the system, h_u represents an external magnetic field acting as a local bias on node u , and $\mathcal{Z}$ is the intractable partition function required for normalization.

From a statistical learning perspective, this model is equivalent to an Energy-Based Model, where the symmetric interaction matrix J defines the network topology and parameterizes the local spatial autocorrelation structures.

In the context of geostatistical spatial modeling, such as Kriging, spatial dependencies are explicitly modeled via parameterized covariance functions; within the Ising MRF framework, the interaction terms J_{uv} serve an identical conceptual purpose by dictating the local spatial autocorrelation. A positive coupling ($J_{uv} > 0$) enforces a positive spatial autocorrelation (a ferromagnetic assumption where neighbors tend to align), whereas a negative coupling ($J_{uv} < 0$) enforces a negative spatial autocorrelation (antiferromagnetic structure).

The primary computational challenge for Bayesian inference lies in the partition function $\mathcal{Z}$, which requires an intractable summation over all 2^n possible configurations. To bypass this bottleneck, Mean-Field Variational Inference approximates the coupled joint distribution $p(x)$ by restricting the optimization to the family of fully factorized independent product distributions $\mathcal{F}_{MF}$ [5]:

$$\mathcal{F}_{MF} := \left\{ q(x) = \prod_{u \in V} q_u(x_u) \right\}. \quad (5)$$

For the foundational two-node spatial network ($n = 2$) analyzed in this study, the Mean-Field assumption removes the explicit dependency induced by the coupling J_{12} in the pairwise target distribution, yielding the factorized approximation $q(x) = q_1(x_1)q_2(x_2)$. The local spatial interaction is therefore no longer captured explicitly through the joint probability measure, but is instead preserved implicitly through the coupled expected values of the independent marginals, reflecting the mutual influence of neighboring nodes during the optimization process.

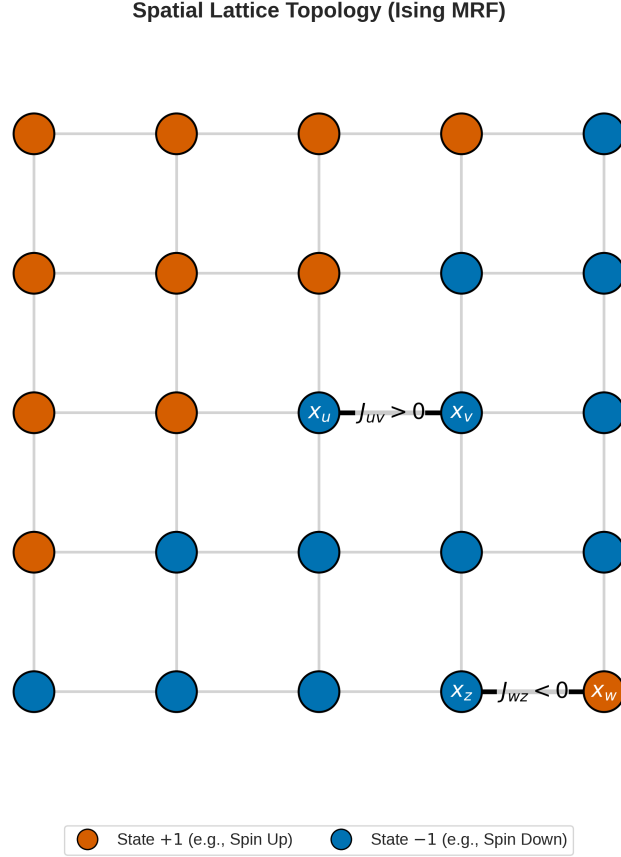


Figure 1: Spatial lattice topology of a two-dimensional Ising Markov Random Field (MRF). The discrete nodes $x_u \in \{-1, 1\}$ represent binary spatial variables, and edges encode pairwise interactions J_{uv} between neighboring nodes. Positive interaction ($J_{uv} > 0$) favors aligned states (ferromagnetic coupling), while negative interactions ($J_{uv} < 0$) favors opposing states (antiferromagnetic coupling). The resulting lattice defines the local spatial dependencies that are subsequently approximated by the mean-field variational model.

5. The ELBO Geometry and CAVI Derivation

Variational inference reframes the infeasible computation of the posterior distribution into a deterministic optimization problem by finding a distribution $q(x)$ that minimizes the Kullback-Leibler (KL) divergence to the posterior. Because direct minimization is computationally infeasible, the optimization is equivalently cast as maximizing the Evidence Lower Bound (ELBO). For a generic target distribution $p(x) = \tilde{p}(x)/\mathcal{Z}$, where $\tilde{p}(x)$ represents the unnormalized probability measure, the ELBO is formulated as:

$$\text{ELBO}(q) = \mathbb{E}_q[\log \tilde{p}(x)] - \mathbb{E}_q[\log q(x)]$$

where the first term represents the expected unnormalized log-joint probability and the second term is the negative entropy of the variational distribution. For the Ising Markov Random Field (MRF), the target distribution is defined in terms of the unnormalized log-probability $\Psi(x)$ as $p(x) = \frac{1}{\mathcal{Z}} \exp\{\Psi(x)\}$.

Thus, the objective function simplifies to:

$$\text{ELBO}(q) = \mathbb{E}_q[\Psi(x)] - \mathbb{E}_q[\log q(x)]. \quad (6)$$

To construct the explicit geometry of the ELBO for the two-node Ising model, we impose the Mean-Field assumption, which restricts the variational family to fully factorized distributions, $q(x) = q_1(x_1)q_2(x_2)$, where $x_1, x_2 \in \{-1, 1\}$ represent the binary spatial nodes. We parameterize their marginal distributions by the probability of occupying the +1 state:

$$\begin{aligned} \zeta = q_1(x_1 = 1), & \implies q_1(x_1 = -1) = 1 - \zeta, \\ \xi = q_2(x_2 = 1), & \implies q_2(x_2 = -1) = 1 - \xi. \end{aligned} \quad (7)$$

Under this parameterization, the expected value of the first node is computed as:

$$\mathbb{E}_{q_1}[x_1] = (1)(\zeta) + (-1)(1 - \zeta) = 2\zeta - 1. \quad (8)$$

An identical expansion for the second node yields $\mathbb{E}_{q_2}[x_2] = 2\xi - 1$.

We now expand the two components of the ELBO step by step. First, we compute the expected energy (the *Model Fit* term). For the two-node spatial network, the negative Hamiltonian is given by $\Psi(x) = \beta J_{12}x_1x_2 + \beta h_1x_1 + \beta h_2x_2$. Taking the expectation with respect to the fully factorized joint distribution $q_1(x_1)q_2(x_2)$, we leverage the linearity of expectation and the independence of the marginals:

$$\begin{aligned} \mathbb{E}_q[\Psi(x)] &= \mathbb{E}_{q_1, q_2}[\beta J_{12}x_1x_2 + \beta h_1x_1 + \beta h_2x_2] \\ &= \beta J_{12}\mathbb{E}_{q_1}[x_1]\mathbb{E}_{q_2}[x_2] + \beta h_1\mathbb{E}_{q_1}[x_1] + \beta h_2\mathbb{E}_{q_2}[x_2] \\ &= \beta J_{12}(2\zeta - 1)(2\xi - 1) + \beta h_1(2\zeta - 1) + \beta h_2(2\xi - 1). \end{aligned} \quad (9)$$

Next, we expand the expectation of the log-variational distribution (the negative *Entropy* term). Because $q(x)$ is factorized, the logarithm splits additively:

$$\begin{aligned} -\mathbb{E}_q[\log q(x)] &= - \sum_{x_1 \in \{-1, 1\}} \sum_{x_2 \in \{-1, 1\}} q_1(x_1)q_2(x_2)[\log q_1(x_1) + \log q_2(x_2)] \\ &= - \sum_{x_1} q_1(x_1) \log q_1(x_1) \sum_{x_2} q_2(x_2) - \sum_{x_2} q_2(x_2) \log q_2(x_2) \sum_{x_1} q_1(x_1) \\ &= -[\zeta \log \zeta + (1 - \zeta) \log(1 - \zeta)] - [\xi \log \xi + (1 - \xi) \log(1 - \xi)] \\ &= \mathcal{H}(\zeta) + \mathcal{H}(\xi), \end{aligned} \quad (10)$$

where $\mathcal{H}(\cdot)$ defines the standard binary entropy function. Combining Equation (9) and Equation (10) yields the fully expanded mathematical form of the objective function:

$$\text{ELBO}(\zeta, \xi) = \underbrace{\beta J_{12}(2\zeta - 1)(2\xi - 1) + \beta h_1(2\zeta - 1) + \beta h_2(2\xi - 1)}_{\text{Model Fit}} + \underbrace{\mathcal{H}(\zeta) + \mathcal{H}(\xi)}_{\text{Regularization}}. \quad (11)$$

The geometry of the ELBO reveals a fundamental machine learning tradeoff: the *Model Fit* term drives the variational parameters ζ and ξ toward states that minimize the empirical loss, favoring configurations aligned with the spatial interactions and, in the strong coupling regime, pushing the parameters toward the deterministic boundaries $\{0, 1\}$.

In contrast, the entropy acts as a *Regularization* term, penalizing deterministic solutions and favoring the maximally uncertain state $(1/2, 1/2)$. The hyperparameter β controls the relative strength of the *Model Fit* and of the *Regularization* terms.

To maximize this objective, the Coordinate Ascent Variational Inference (CAVI) algorithm iteratively optimizes the ELBO with respect to one coordinate at a time. The optimal update rule for a generic node u in the Mean-Field framework is given by:

$$q_u^*(x_u) \propto \exp\{\mathbb{E}_{-u}[\Psi(x)]\}, \quad (12)$$

where $\mathbb{E}_{-u}$ denotes the expectation taken over all marginal distributions except q_u .

For node 1, we must compute the expectation of the energy function with respect to the current state of node 2 (parameterized by ξ_k). We isolate the components of $\Psi(x_1, x_2)$ and compute the partial expectation:

$$\begin{aligned}\mathbb{E}_{q_2}[\Psi(x_1, x_2)] &= \mathbb{E}_{q_2}[\beta J_{12}x_1x_2 + \beta h_1x_1 + \beta h_2x_2] \\ &= \beta J_{12}x_1\mathbb{E}_{q_2}[x_2] + \beta h_1x_1 + \beta h_2\mathbb{E}_{q_2}[x_2] \\ &= \beta J_{12}x_1(2\xi_k - 1) + \beta h_1x_1 + \beta h_2(2\xi_k - 1).\end{aligned}\quad (13)$$

To find the probability distribution for node 1, we evaluate the exponential of this expectation at the two possible states, $x_1 = 1$ and $x_1 = -1$:

$$\text{For } x_1 = 1: \quad \exp\{\beta J_{12}(2\xi_k - 1) + \beta h_1 + \beta h_2(2\xi_k - 1)\}, \quad (14)$$

$$\text{For } x_1 = -1: \quad \exp\{-\beta J_{12}(2\xi_k - 1) - \beta h_1 + \beta h_2(2\xi_k - 1)\}. \quad (15)$$

The variational update $\zeta_{k+1} = q_1^*(x_1 = 1)$ is computed by normalizing (14) by the sum of evaluations at both possible states. Because the term $\beta h_2(2\xi_k - 1)$ does not depend on x_1 , it factors out as a constant scalar and cancels from the expression:

$$\zeta_{k+1} = \frac{\exp\{\beta J_{12}(2\xi_k - 1) + \beta h_1\} \cdot e^{\beta h_2(2\xi_k - 1)}}{\left[\exp\{\beta J_{12}(2\xi_k - 1) + \beta h_1\} + \exp\{-\beta J_{12}(2\xi_k - 1) - \beta h_1\}\right] \cdot e^{\beta h_2(2\xi_k - 1)}}. \quad (16)$$

Canceling the common exponential scalar, we are left with the isolated structural dependence:

$$\zeta_{k+1} = \frac{\exp\{\beta J_{12}(2\xi_k - 1) + \beta h_1\}}{\exp\{\beta J_{12}(2\xi_k - 1) + \beta h_1\} + \exp\{-\beta J_{12}(2\xi_k - 1) - \beta h_1\}}. \quad (17)$$

Dividing both the numerator and the denominator by $\exp\{\beta J_{12}(2\xi_k - 1) + \beta h_1\}$ yields the standard logistic form:

$$\zeta_{k+1} = \frac{1}{1 + \exp\{-2\beta J_{12}(2\xi_k - 1) - 2\beta h_1\}}. \quad (18)$$

Assuming a standardized spatial interaction ($J_{12} = 1$) and zero external magnetic field ($h_1 = h_2 = 0$), and defining the standard logistic function as $\sigma(u) = (1 + e^{-u})^{-1}$, the sequence formulation collapses into a discrete time recursive system:

$$\zeta_{k+1} = \sigma(4\beta\xi_k - 2\beta), \quad \xi_{k+1} = \sigma(4\beta\zeta_{k+1} - 2\beta). \quad (19)$$

Theorem 1 (Monotonic Convergence of Sequential CAVI):

Let the variational family be restricted to $\mathcal{F}_{MF}$. The sequential Coordinate Ascent Variational Inference (CAVI) update for a single marginal $q_u(x_u)$, obtained by computing (12), generates a monotonically non-decreasing sequence with respect to the Evidence Lower Bound

Proof: By definition, the ELBO can be additively decomposed with respect to a single marginal $q_u(x_u)$ while holding all other marginal distributions $q_{v \neq u}(x_v)$ fixed. Up to additive constants independent of q_u , the objective function simplifies to:

$$\text{ELBO}(q_u) = \mathbb{E}_{q_u}[\mathbb{E}_{-u}[\Psi(x)]] - \mathbb{E}_{q_u}[\log q_u(x_u)] + \text{const}$$

This restricted functional is strictly concave with respect to q_u . The functional derivative of this Lagrangian with respect to q_u , subject to the probability normalization constraint $\sum_{x_u} q_u(x_u) = 1$, yields a unique global maximum for this one dimensional slice of the parameter space. Because the sequential CAVI step replaces q_u with this analytical maximum, the overall ELBO is mathematically guaranteed to either increase or remain unchanged if the coordinate is already at a stationary point. Thus, the sequential coordinate wise updates construct a monotonically non decreasing sequence. ■

Equation (19) exhibits a recurrent structure analogous to that of a single layer Recurrent Neural Network (RNN). The sequential CAVI update can be interpreted as a forward pass through this recurrent architecture. The expected state of the spatial neighbor, $(2\xi_k - 1)$, acts as the network’s input signal, while the spatial interaction matrix J_{12} operates as the recurrent weight matrix. Finally, the scaled inverse temperature 2β acts as the network’s gain parameter, governing the steepness of the sigmoid activation function and thus the gain of the nonlinear response. Consequently, mean-field inference on the pairwise MRF admits an interpretation as a discrete time dynamical system, whose iterative updates evolve toward a stationary point of the variational objective.

6. Algorithmic Implementation

The theoretical update equations derived in Section 5 can be translated into computational practice through two distinct algorithmic paradigms. The standard approach to Coordinate Ascent Variational Inference (CAVI) employs a sequential, Gauss-Seidel style update scheme. In this paradigm, each variational parameter is updated and immediately utilized in the subsequent coordinate calculations within the same iteration cycle. This immediate feedback mechanism ensures that each step maximizes the objective with respect to the most current network state, thereby rigorously preserving the monotonic increase of the Evidence Lower Bound (ELBO) [5].

Algorithm 1 Sequential CAVI for two-Node Ising Model

Require: Inverse temperature β , Tolerance ϵ

Ensure: Optimal variational parameters ζ, ξ

```

1: Initialize  $\zeta^{(0)} \in (0, 1)$  and  $\xi^{(0)} \in (0, 1)$  randomly
2:  $k \leftarrow 0$ 
3:  $error \leftarrow \infty$ 
4: while  $error > \epsilon$  do
5:   // Update Node 1 using the previous state of Node 2
6:    $\zeta^{(k+1)} \leftarrow \sigma(4\beta\xi^{(k)} - 2\beta)$ 
7:   // Update Node 2 immediately using the newly computed state of Node 1
8:    $\xi^{(k+1)} \leftarrow \sigma(4\beta\zeta^{(k+1)} - 2\beta)$ 
9:   // Evaluate convergence
10:   $error \leftarrow \max(|\zeta^{(k+1)} - \zeta^{(k)}|, |\xi^{(k+1)} - \xi^{(k)}|)$ 
11:   $k \leftarrow k + 1$ 
12: end while
13: return  $\zeta^{(k)}, \xi^{(k)}$ 
```

Alternatively, the optimization can be structured using a parallel, Jacobi-style block update scheme. In this computational paradigm, all spatial nodes compute their expectations simultaneously, relying only on the global network state from the previous time step (k). While this approach is computationally advantageous for modern Machine Learning frameworks due to its natural compatibility with vectorized tensor operations and highly parallelized computing architectures, the simultaneous evaluation removes the strict coordinate-wise monotonic guarantees of the sequential optimization scheme.

Algorithm 2 Parallel (Block) CAVI for two-Node Ising Model

Require: Inverse temperature β , Tolerance ϵ

Ensure: Optimal variational parameters ζ, ξ

```
1: Initialize  $\zeta^{(0)} \in (0, 1)$  and  $\xi^{(0)} \in (0, 1)$  randomly
2:  $k \leftarrow 0$ 
3:  $error \leftarrow \infty$ 
4: while  $error > \epsilon$  do
5:   //Simultaneously compute new states using ONLY the previous iteration's values
6:    $\zeta_{new} \leftarrow \sigma(4\beta\xi^{(k)} - 2\beta)$ 
7:    $\xi_{new} \leftarrow \sigma(4\beta\zeta^{(k)} - 2\beta)$ 
8:   //Apply the block update
9:    $\zeta^{(k+1)} \leftarrow \zeta_{new}$ 
10:   $\xi^{(k+1)} \leftarrow \xi_{new}$ 
11:  //Evaluate convergence
12:   $error \leftarrow \max(|\zeta^{(k+1)} - \zeta^{(k)}|, |\xi^{(k+1)} - \xi^{(k)}|)$ 
13:   $k \leftarrow k + 1$ 
14: end while
15: return  $\zeta^{(k)}, \xi^{(k)}$ 
```

Mathematically, these two computational paradigms define different dynamical mappings over the parameter space. Let $m_k = (\zeta_k, \xi_k)$ represent the joint spatial state at iteration k . Assuming a standard logistic function $\sigma(u) = (1 + e^{-u})^{-1}$, the parallel update applies the non linear operator simultaneously across all dimensions, defining the map:

$$T_{\text{parallel}} : (\zeta_k, \xi_k) \mapsto \left(\sigma(4\beta\xi_k - 2\beta), \sigma(4\beta\zeta_k - 2\beta) \right). \quad (20)$$

Conversely, the sequential update immediately injects the updated coordinate into the subsequent evaluation, effectively composing the functions within a single iterative step:

$$T_{\text{seq}} : (\zeta_k, \xi_k) \mapsto \left(\sigma(4\beta\xi_k - 2\beta), \sigma(4\beta\sigma(4\beta\xi_k - 2\beta) - 2\beta) \right). \quad (21)$$

This structural distinction explains their different dynamical behaviors. As established in Theorem 1, the sequential map T_{seq} guarantees a monotonic increase in the objective function, $\text{ELBO}(m_{k+1}) \geq \text{ELBO}(m_k)$, meaning that the negative ELBO acts as a strict Lyapunov function for the discrete time dynamical system. This mathematical property prevents sustained non convergent behaviors, such as limit cycles or chaotic orbits, ensuring that the sequential trajectory must eventually terminate at a stationary point (even if it is a suboptimal local mode). By contrast, the simultaneous evaluation in the parallel map T_{parallel} breaks this coordinate-wise monotonic guarantee. Lacking a strict Lyapunov function, the parallel system is free to exhibit generic non convergent dynamical trajectories when the underlying fixed point becomes unstable.

7. Dynamical Systems Analysis of the Loss Landscape

To rigorously evaluate the convergence guarantees of the Coordinate Ascent Variational Inference (CAVI) algorithm, we analyze its update equations as an iterated discrete time dynamical system. By evaluating the local stability of the system's stationary points, we can map the topology of the Evidence Lower Bound (ELBO) and formalize the thresholds where algorithmic failures occur.

For the two-node Ising model, assuming a standardized spatial interaction ($J_{12} = 1$) and zero external field ($h_1 = h_2 = 0$), the decoupled sequential CAVI update for a single coordinate reduces to the one dimensional map $x_{k+1} = f(x_k)$, where $f(x) = \sigma(4\beta x - 2\beta)$ and $\sigma(u) = (1 + e^{-u})^{-1}$. The stationary points of the ELBO correspond exactly to the fixed points of this map, satisfying $x_* = f(x_*)$. To

determine whether the coordinate descent algorithm will converge to a given fixed point, we evaluate the system's derivative at the symmetric stationary point $x_* = 1/2$. Using the chain rule and the standard derivative of the logistic function, the derivative evaluated at this symmetric fixed point yields:

$$f'(1/2) = 4\beta\sigma(0)(1 - \sigma(0)) = 4\beta \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \beta. \quad (22)$$

Theorem 2 (Bifurcation Threshold of the MFVI Loss Landscape):

For the two-node Ising model under Mean-Field Variational Inference, the optimization landscape possesses a locally asymptotically stable fixed point at the symmetric stationary point $(1/2, 1/2)$ if and only if $\beta \leq 1$. For $\beta > 1$, the symmetric stationary point becomes an unstable hyperbolic fixed point following a supercritical pitchfork bifurcation at $\beta = 1$, coexisting with two symmetric stable fixed points corresponding to the global maxima of the ELBO.

Proof: The optimization dynamics of the CAVI algorithm applied to the two-node Ising model can be decoupled into a one dimensional iterated discrete dynamical system governed by the map $x_{k+1} = f(x_k) = \sigma(4\beta x_k - 2\beta)$. By the Hartman-Grobman theorem, the local asymptotic stability of a fixed point x_* is topologically equivalent to the behavior of its linear approximation, provided the fixed point is hyperbolic (i.e., its derivative satisfies $|f'(x_*)| \neq 1$). Evaluating the derivative of the system at the symmetric fixed point $x_* = 1/2$ yields $f'(1/2) = \beta$, which defines the sole eigenvalue $\lambda = \beta$ of the linearized system.

For $\beta < 1$ (the Dobrushin regime), the eigenvalue is strictly bounded inside the unit circle ($|\lambda| < 1$). Therefore, the fixed point is locally asymptotically stable. In this regime, $(1/2, 1/2)$ is the unique global maximizer of the ELBO, and the CAVI dynamics converge to it from every initialization.

At the critical threshold $\beta = 1$, the eigenvalue equals exactly 1, breaking hyperbolicity and indicating a local bifurcation. To characterize the local structure of this bifurcation, let $y = x - \frac{1}{2}$ represent the local perturbation around the symmetric fixed point. A Taylor expansion of the map $f(x)$ around this critical configuration yields:

$$f\left(\frac{1}{2} + y\right) - \frac{1}{2} = \beta y - \frac{4\beta^3}{3}y^3 + \mathcal{O}(y^5). \quad (23)$$

At the bifurcation threshold $\beta = 1$, the linear term becomes neutral ($\lambda = 1$), while the leading non-linear term is cubic and stabilizing, as its coefficient is strictly negative ($-\frac{4}{3} < 0$).

For $\beta > 1$, the eigenvalue strictly exceeds unity ($\lambda = \beta > 1$). Consequently, the linear term becomes expanding, while the negative cubic term provides the nonlinear saturation that generates two new symmetry-related non-zero fixed points at $\pm y^*$. This confirms the supercritical pitchfork structure. At the optimization level, these two stable fixed points correspond to the two symmetry-broken local maxima of the ELBO, toward which the sequential CAVI dynamics converge depending on the initialization. ■

Dynamical systems theory states that a fixed point is locally asymptotically stable if the magnitude of its derivative is strictly less than 1. Equation (22) formalizes the behavior within the Dobrushin regime ($|\beta| < 1$). In this high temperature state, the entropy regularization dominates the ELBO, ensuring that the objective function remains globally strictly concave. The CAVI algorithm acts as a strict contraction mapping, guaranteeing monotonic convergence to the globally optimal true posterior mean $(1/2, 1/2)$ regardless of the initialization.

In contrast, as the inverse temperature increases and crosses the critical threshold $\beta > 1$, the eigenvalue exceeds unity, triggering a topological shift. The symmetric stationary point $(1/2, 1/2)$ transforms into an unstable hyperbolic saddle. Simultaneously, two symmetric, suboptimal local maxima emerge. Because the true mean is an unstable fixed point, the sequential CAVI algorithm breaks symmetry and converges to one of the biased local modes depending on its initialization.

This bifurcation behavior is not an isolated physical quirk of the Ising model, but rather a systemic algorithmic vulnerability inherent to the Mean-Field assumption. Recent literature demonstrates that identical pitchfork bifurcations characterize standard predictive models in machine learning, such as

Topic Models and Stochastic Block Models, driving variational estimators to converge to uncorrelated, suboptimal local modes when traversing non convex loss manifolds [3].

Furthermore, attempting to accelerate the optimization via parallelization introduces even more severe topological anomalies. While sequential coordinate descent updates immediately incorporate newly computed marginals, parallel (block) CAVI updates both variables simultaneously based on the previous time step. At the strongly antiferromagnetic critical threshold $\beta = -1$, the eigenvalue hits -1 . While sequential CAVI undergoes another pitchfork bifurcation, the parallelized system transitions into a bounded, non convergent oscillatory regime. The parallel updates over correct, causing the spatial parameters to oscillate. While establishing the existence of an asymptotically stable limit cycle requires a higher order analysis of the second iterate $f^{(2)}(x)$, the strict instability condition $f'(x_*) < -1$ guarantees that the symmetric state is repelling. The optimization trajectory is driven into non convergent, oscillatory dynamics, preventing the parallel algorithm from ever reaching a stationary point.

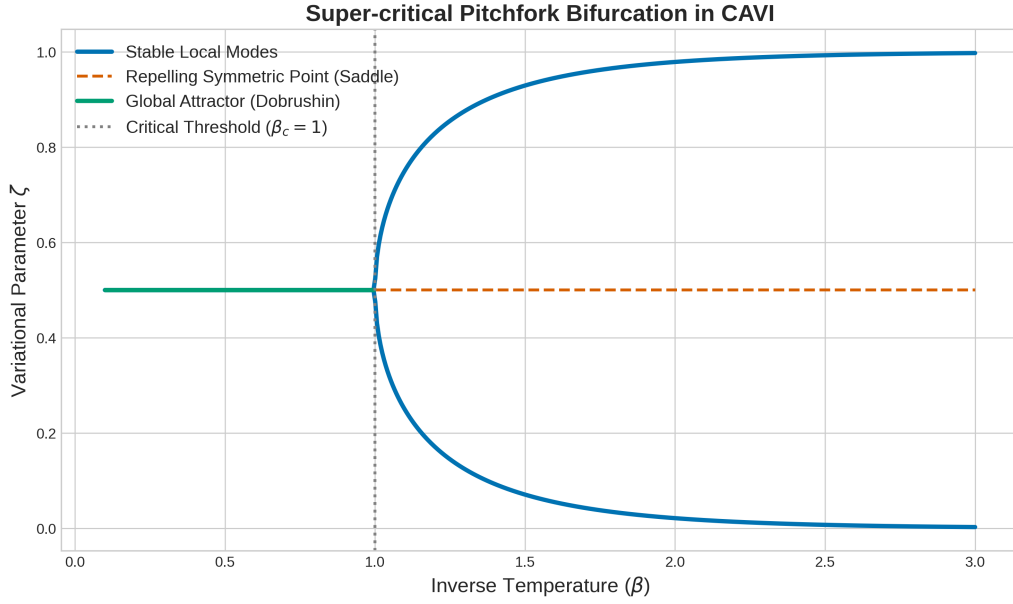


Figure 2: Bifurcation diagram mapping the topological evolution of the ELBO stationary points as a function of the inverse temperature parameter β . Within the Dobrushin regime ($|\beta| \leq 1$), the optimization landscape is globally strictly concave, displaying a single stable fixed point at $(1/2, 1/2)$, which acts as a global attractor for the CAVI algorithm. At the critical threshold $\beta = 1$, the eigenvalue of the linearized system exceeds unity, triggering a supercritical pitchfork bifurcation. Beyond this threshold, the true mean becomes an unstable saddle point, and the manifold develops two symmetric, suboptimal stable branches that attract the optimization trajectory depending on the initial conditions.

This diagram visualizes the algorithmic danger of applying mean-field assumptions to strongly correlated networks: outside the convex regime, the optimization algorithm is deterministically trapped in biased local modes, breaking symmetry depending entirely on its initialization coordinates.

8. Critical Limitations: Network Scalability and Regularization

The bifurcation analysis presented in Section 7 relies on two restrictive assumptions: a simplified two-node topology and the absence of an external magnetic field ($h = 0$). To generalize these findings to applied statistical learning, we evaluate how relaxing these mathematical boundaries alters the geometry of the ELBO.

Scaling the system from a two-node configuration to an N -node lattice graph exponentially increases the complexity of the non concave manifold. Rather than decoupling into tractable one dimensional maps, the CAVI algorithm evolves as a coupled N -dimensional dynamical system. In geostatistical contexts, such as Kriging estimation, this is analogous to incorporating dense spatial autocorrelation matrices across a vast geographic grid. Outside the Dobrushin regime ($|\beta| > 1$), the pitchfork bifurcation still occurs; however, instead of splitting into merely two symmetric modes, the number of local optima in the ELBO can grow exponentially. The interacting variables group into localized ferromagnetic clusters or alternating antiferromagnetic configurations. While the symmetric stationary point remains a repeller, the probability of the coordinate descent algorithm successfully navigating this highly non concave landscape to locate the global maximum decreases exponentially as N scales.

Conversely, introducing a non zero external magnetic field ($h \neq 0$) alters the topological nature of the bifurcations. The ideal pitchfork bifurcation requires an energy function with Z_2 symmetry (even symmetry). A non zero external field breaks this symmetry (i.e., $f_{xx} \neq 0$ at the critical point), transforming the pitchfork bifurcation into an imperfect saddle node bifurcation.

In the framework of machine learning, the external field h acts as an informative Bayesian prior or a node specific bias. This term introduces asymmetry into the loss landscape. Under a weak prior, the manifold retains multiple local optima, but one basin of attraction becomes deeper and wider than the other, biasing the CAVI convergence trajectory toward a preferred mode. If this informative prior h is sufficiently strong, it overpowers the pairwise spatial interactions J_{uv} , eliminating the suboptimal mode and restoring the ELBO to a globally concave state.

Thus, a strong Bayesian prior guarantees algorithmic convergence by enforcing a topology that resembles the high temperature contraction regime, even at low temperatures (large β).

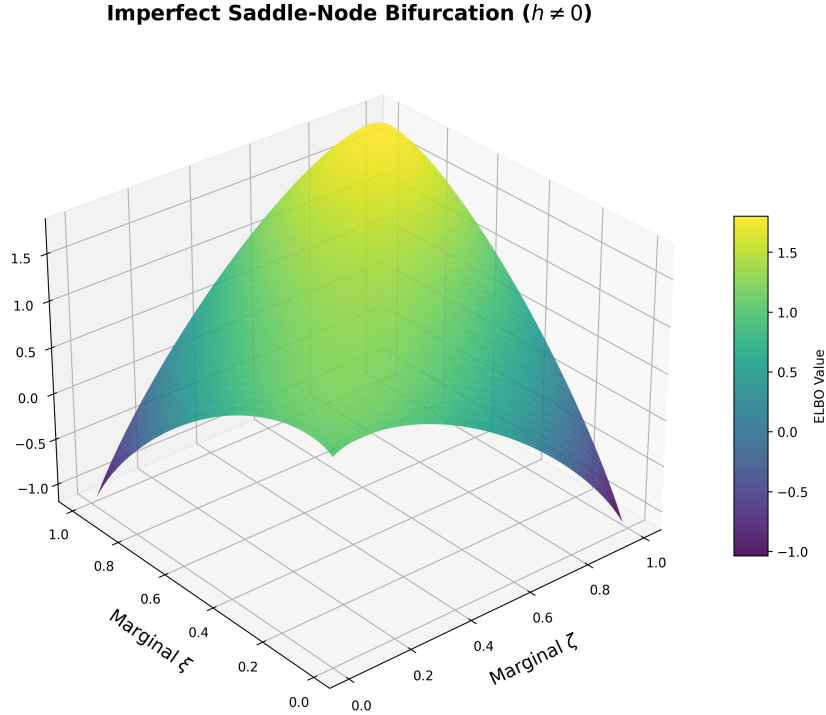


Figure 3: Three-dimensional surface visualization of the non concave ELBO geometry under a non zero external magnetic field ($h \neq 0$). In the standard Ising formulation with zero external field, the energy function possesses a global Z_2 symmetry, ensuring a symmetric supercritical pitchfork bifurcation. Introducing an external field breaks this symmetry, transforming the topological transition into an imperfect saddle node bifurcation. This parameterization introduces an asymmetric bias, causing one basin of attraction to become energetically and probabilistically favored over the other, as reflected by the dominant peak in the surface. A sufficiently strong prior suppresses the suboptimal local modes, restoring global concavity and ensuring monotonic convergence to a unique stationary point.

9. Expanding the Manifold: Auxiliary Variables

The bifurcation analysis presented in the preceding sections establishes a fundamental limitation of standard Mean-Field Variational Inference: when applied to strongly correlated spatial networks, the rigid geometric constraints of the factorized family restrict the Evidence Lower Bound (ELBO) into a non concave topology. To preserve the computational efficiency of the CAVI algorithm while avoiding these dynamical issues, the optimization manifold itself must be modified.

In statistical physics, a canonical solution to the poor mixing of the Ising model at low temperatures is the Edwards-Sokal coupling. Rather than modeling the node states x in isolation, this approach introduces an auxiliary bond (or edge) variable $\omega_{uv} \in \{0, 1\}$ between any two interacting nodes. The binary state of ω_{uv} dictates whether the spatial interaction J_{uv} is active or inactive, constructing an augmented joint probability measure $\mu(x, \omega)$ over both vertices and edges. Marginalizing out the auxiliary bond variables ω , we exactly recover the original Ising Markov Random Field (MRF).

In the paradigm of modern statistical learning, the Edwards-Sokal coupling can be formalized as an instance of Data Augmentation or to the introduction of Auxiliary Latent Variables [1]. By defining an augmented variational family $q(x, \omega)$ and maximizing the augmented ELBO,

$$\text{ELBO}_{\text{aug}}(q) = \mathbb{E}_{q(x, \omega)}[\log \mu(x, \omega)] - \mathbb{E}_{q(x, \omega)}[\log q(x, \omega)], \quad (24)$$

the optimization problem is lifted from the lower dimensional space of the original parameters into a higher dimensional augmented manifold. This extra dimensionality provides the coordinate ascent algorithm with additional degrees of freedom. The structural dependencies induced by the pairwise interactions J_{uv} , which previously triggered supercritical pitchfork bifurcations, are decomposed through the latent space. The expanded parameter space alters the optimization landscape topology, allowing the CAVI algorithm to circumvent the saddle points and the repellers that trap the standard mean-field approach. Thus, the convergence domain expands, providing a pathway to circumvent local modes and mitigate the loss of stability well outside the boundaries of the Dobrushin regime.

10. Conclusion

This report demonstrates that analyzing Variational Inference exclusively through the lens of statistical estimation limits the understanding of its underlying optimization mechanics. By reframing the CAVI algorithm as a discrete dynamical system, we establish a correspondence between physical phase transitions and algorithmic instabilities. Using the Ising model as a prototypical spatial MRF, conceptually mirroring the spatial autocorrelation structures foundational to geostatistical estimation techniques like Kriging, we mapped the exact geometric boundaries of the ELBO.

We showed analytically that, while the mean-field assumption provides a globally concave, rapidly converging optimization problem within the Dobrushin regime, crossing this critical temperature threshold alters the topology of the ELBO. The symmetric stationary point becomes an unstable hyperbolic saddle, driving sequential coordinate descent to break symmetry and converge to biased local modes, while simultaneously causing parallel block updates to exhibit non convergent, oscillatory dynamics. Ultimately, successfully deploying Variational Inference in strongly coupled spatial models requires a deep understanding of this underlying geometry. When deterministic optimization reaches its dynamical limits, we must leverage topological expansions, such as auxiliary latent variables, to unfold the non convex manifold and mitigate the loss of algorithmic stability.

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